

A Deep Learning Approach for the Coconut Tree Disease Detection

Presented by

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Outline

- Introduction
- Objective
- Background Study
- Gap Analysis
- Methodology
- Results & Analysis
- Novelty of the Work
- Conclusion
- References

Introduction

- ❑ **The Problem:** Coconut diseases such as Bud Rot, Leaf Rot, Yellow Leaf Disease, Coconut Caterpillar, and Stem Bleeding cause major agricultural losses. Manual disease identification is slow, inconsistent, and highly expertise-dependent.
- ❑ **The Solution:** This project uses deep learning models to classify the disease accurately.
- ❑ **The Gap:** There aren't few diverse coconut datasets available, and most existing models don't explain how they make their predictions.
- ❑ **The Goal:** Identify accurate and lightweight models, and use XAI tools to make the system transparent and reliable for real-world use.



Objective

- ❑ **Develop an accurate coconut disease classifier** using a lightweight transfer-learning model (MobileNetV2, MobileNetV3).
- ❑ **Build a high-quality dataset** of five coconut diseases using real-world images collected from publicly available datasets.
- ❑ **Evaluate model performance** using key metrics: Accuracy, Precision, Recall, and F1-Score, to identify the most efficient and reliable architecture.
- ❑ **Integrate Explainable AI (XAI)** techniques (Grad-CAM, Grad-CAM ++)

Background Study

- ❑ Why coconut disease detection matters?
- ❑ Drawbacks of manual identification
- ❑ Growth of machine vision & AI
- ❑ What previous research achieved
- ❑ Remaining challenges & need for this work

Similar Studies

- ❑ **Deep Learning disease detection:** Many studies use deep learning models such as Faster R-CNN, SSD, YOLO, and various CNN architectures for pest identification, achieving strong performance ranging from 82% to 98% accuracy across different crops.
- ❑ **Use of Lightweight and Mobile-Friendly Models:** Research highlights the effectiveness of efficient models like MobileNet and MobileNetV3 for real-time detection, proving that lightweight architectures can operate reliably in resource-limited field environments.
- ❑ **Real-Time Detection in Agriculture:** Works using Faster R-CNN, SSD, and YOLOv5-S demonstrate that pest detection highlights the importance of reliable classification using deep learning models.

Similar Studies

- ❑ **Explainable AI in Botany:** Recent research introduces Grad-CAM and similar visualization methods to interpret model decisions, making tree disease identification systems more transparent for botanical experts.
- ❑ **Challenges Noted in Previous Work:** Studies often struggle with visual disease, cluttered backgrounds, and inconsistent lighting, emphasizing the need for robust preprocessing and augmentation; issues this project directly addresses.

Comparison Between Existing Works

Author	Year	Title	Methodology	Key Findings
Subbaian et al. [1]	2024	Coconut leaf disease detection	YOLOv4	Achieved 88% F1-score and 85% mAP for leaf disease detection.
Singh et al. [2]	2021	Coconut disease and pest detection	CNN, MobileNet, InceptionResNet V2	Custom CNN achieved 96.94% accuracy.
Nivethitha et al. [5]	2022	Review on coconut disease detection	Deep Learning Review	Summarized CNN-based methods and challenges.
Muthulakshmi et al. [7]	2025	Coconut disease classification	DenseNet201, ResNet50, EfficientNetB0	Achieved up to 99.85% accuracy.
George et al. [8]	2023	Early detection of stem bleeding	Color thresholding	Achieved 80.36% accuracy.

Table 1.1 Comparison Between Existing Works

Comparison Between Existing Works

Author	Year	Title	Methodology	Key Findings
Santhi & Murugan [12]	2025	Cocoscan: Precision diagnostics for coconut leaf disease	InceptionV3, Web App, XAI	Real-time disease diagnosis with XAI-based visual explanations for user trust
Pandian et al. [15]	2022	Plant disease detection using deep CNN	14-layer custom DCNN	Achieved 99.97% accuracy with large augmented dataset across 58 disease classes.
Vidhanaarachchi et al. [21]	2025	Early diagnosis of coconut leaf wilt & caterpillar infestation	CNN, Mask R-CNN, YOLO	Achieved 90–97% disease accuracy and >95% caterpillar detection rate.

Table 1.1 Comparison Between Existing Works

Gap Analysis

- ❑ **Limited studies focus on coconut tree disease.**
- ❑ **Datasets are limited or imbalanced**, lacking real-world variation.
- ❑ **Heavy models dominate research**; lightweight, mobile-friendly models are rarely explored.
- ❑ **XAI is missing in most works**, making predictions hard to interpret.
- ❑ **Minimal multi-model comparison** on a single dataset to identify the best architecture.
- ❑ **Lack of practical systems**, such as deployable apps or user-facing interfaces.
- ❑ **Real-world challenges** like diverse regions, lighting variation, and similar disease are not well addressed.

Methodology

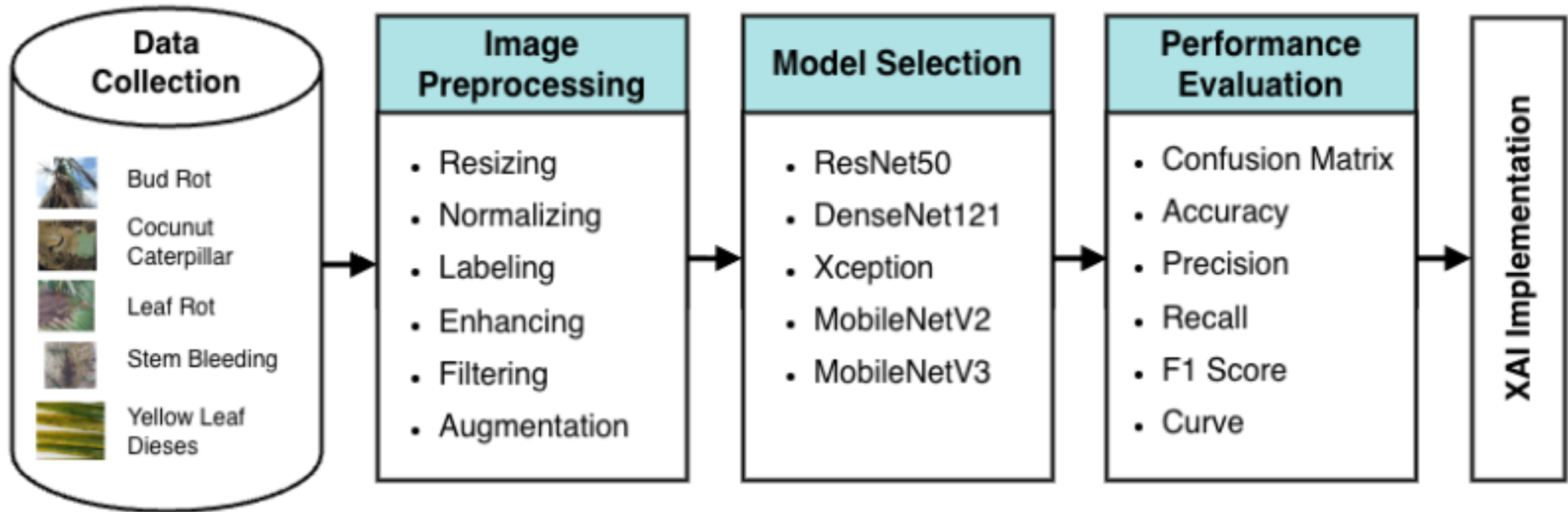


Figure 1.1: Detailed Methodology Diagram

Methodology

Data Collection:

- Data was collected from multiple publicly available online datasets.
- Faced challenges with lighting variability and cluttered backgrounds, addressed through data preprocessing and cleaning.
- Expanded the dataset to **5000** sample images after augmentation.

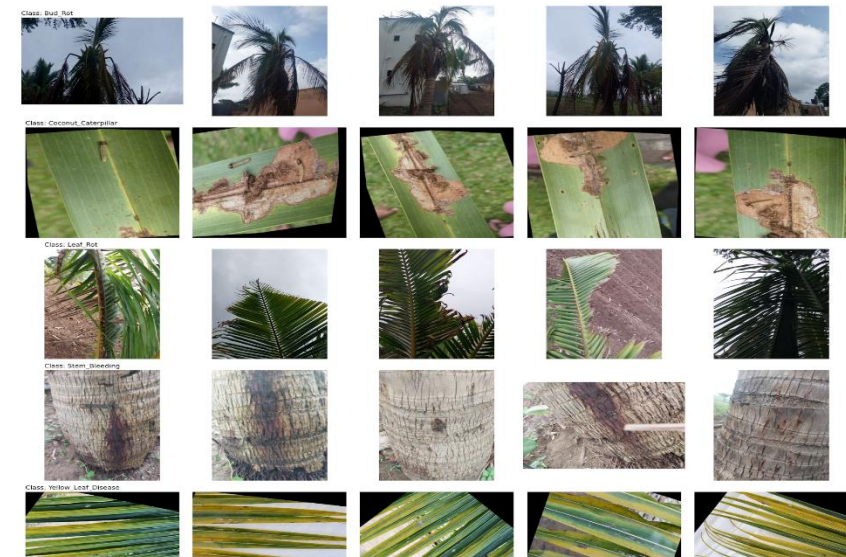


Figure 1.2: Sample images from each class.

Methodology

Image Preprocessing:

- ❖ Resized images to 224 x 224 for uniform input.
- ❖ Applied contrast enhancement and noise reduction.
- ❖ Performed manual labeling for accurate class assignment.

Data Augmentation:

- ❖ Techniques: rotation, width/height shift, zoom ($\leq 20\%$), brightness adjustment, rescaling.
- ❖ Enhanced dataset diversity without altering key features.
- ❖ Expanded dataset to 5000 images evenly across 5 coconut diseases.

Methodology

Data Visualization

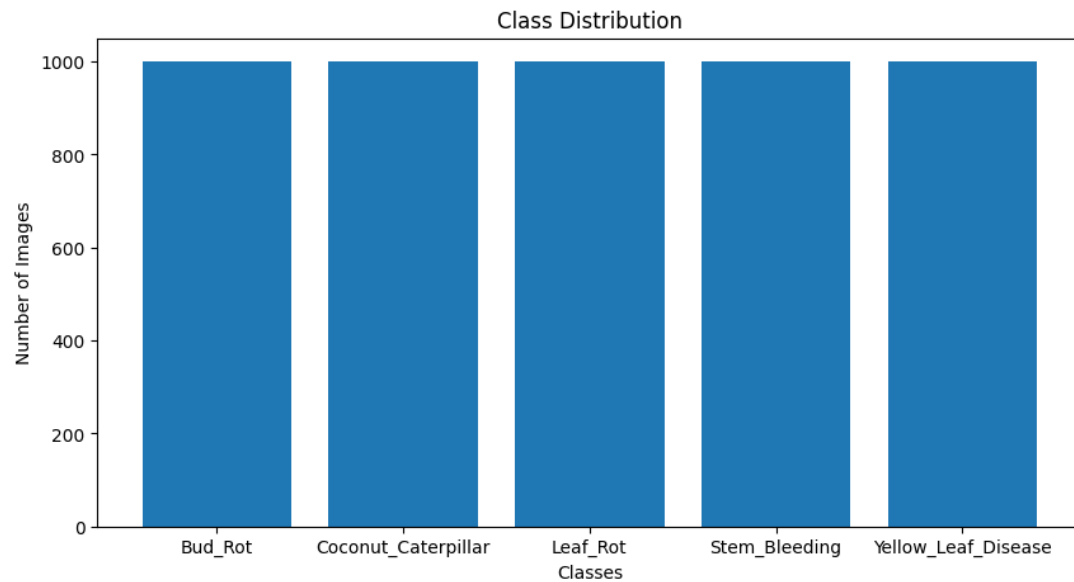


Figure 1.3: Class distribution after data augmentation



Figure 1.4: Sample images from each class (After Augmentation)

Methodology

The research used a systematic research process to come up with a reliable and scalable taxonomic system of the coconut leaf diseases. To begin with, related literature on the application of deep learning in the diagnosis of plant diseases was reviewed[1][2]. It involved CNN architectures[5], mobile-deployable lightweight models, ensemble learning, and Explainable AI technologies such as Grad-CAM[12]. The review allowed us to identify some of the key flaws of prior research, such as not having a balanced set of datasets, mistakenly grouped diseases, and models that are not user-friendly. The given methodology will aim to increase accuracy through fusion models, reduce computational requirements by using lightweight networks, balance the dataset, and provide explainability to strengthen trust in forecasting. Testing the performance, the comparison of various models, and making preparations to use it in real-time are also aspects of the methodology.

Methodology

Model Selection

II) Transfer Learning Models:

i) MobileNetV2

The MobileNetV2 is a small model that can be used to implement high-speed inferences and, therefore, can be deployed on mobile and real-time applications of disease recognition. It's separable convolutions that can be computed depth-wise wise saving on a lot of computations, but offer high levels of accuracy as has been evident in the region of medicinal plant and general object classification [2][36]. MobileNetV2 plays an important role in our study as it is applied to assess the fast and efficient performance of classifications, which is relevant in cases when applications demand low-latency classifications.

Methodology

Model Visualization

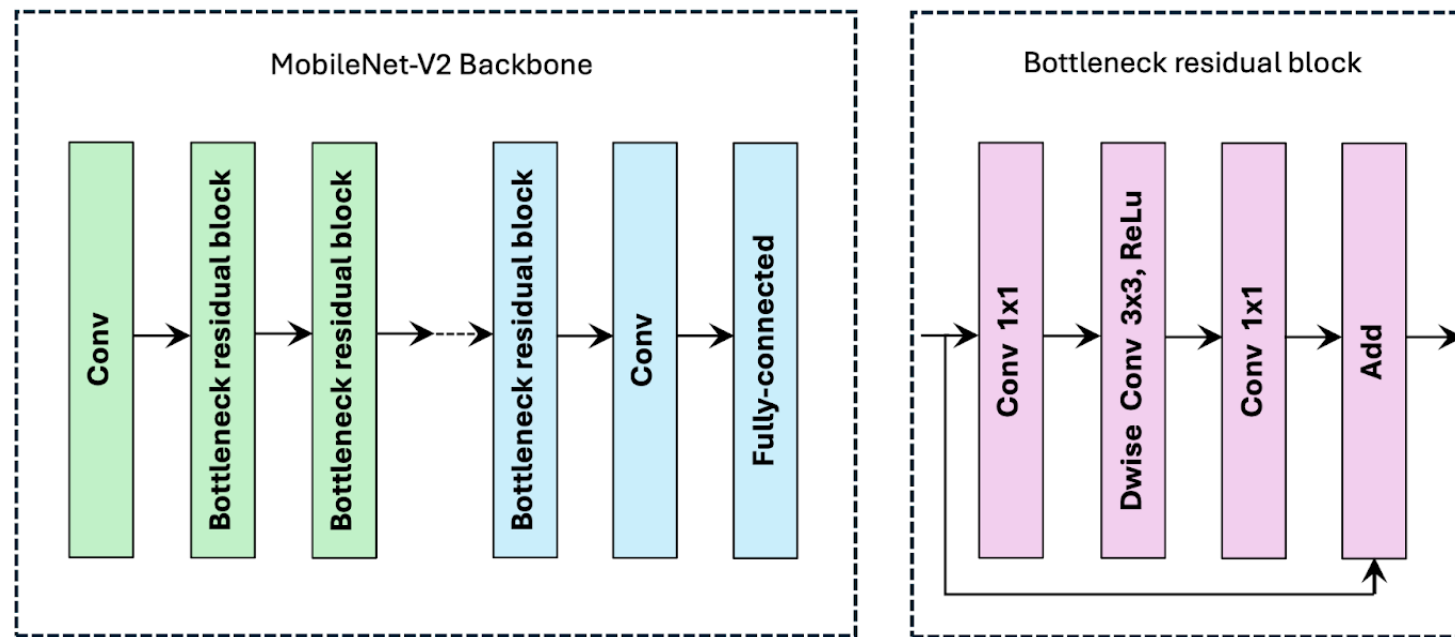


Figure 1.5: MobileNetV2 Architecture Diagram
Mifthauddin et al.2025[36]

Methodology

Model Selection

II) Transfer Learning Models:

ii) MobileNetV3

MobileNetV3 will utilize MobileNetV2, although it includes the neural architecture search (NAS) and squeeze-and-excitation (SE) modules to enhance the network automatically. It is available in two sizes, Small and Large. The two are designed to be as efficient and precise as possible. MobileNetV3 is even more efficient in consuming computer resources than the former and performs better in a variety of benchmark datasets [36].

Methodology

Model Visualization

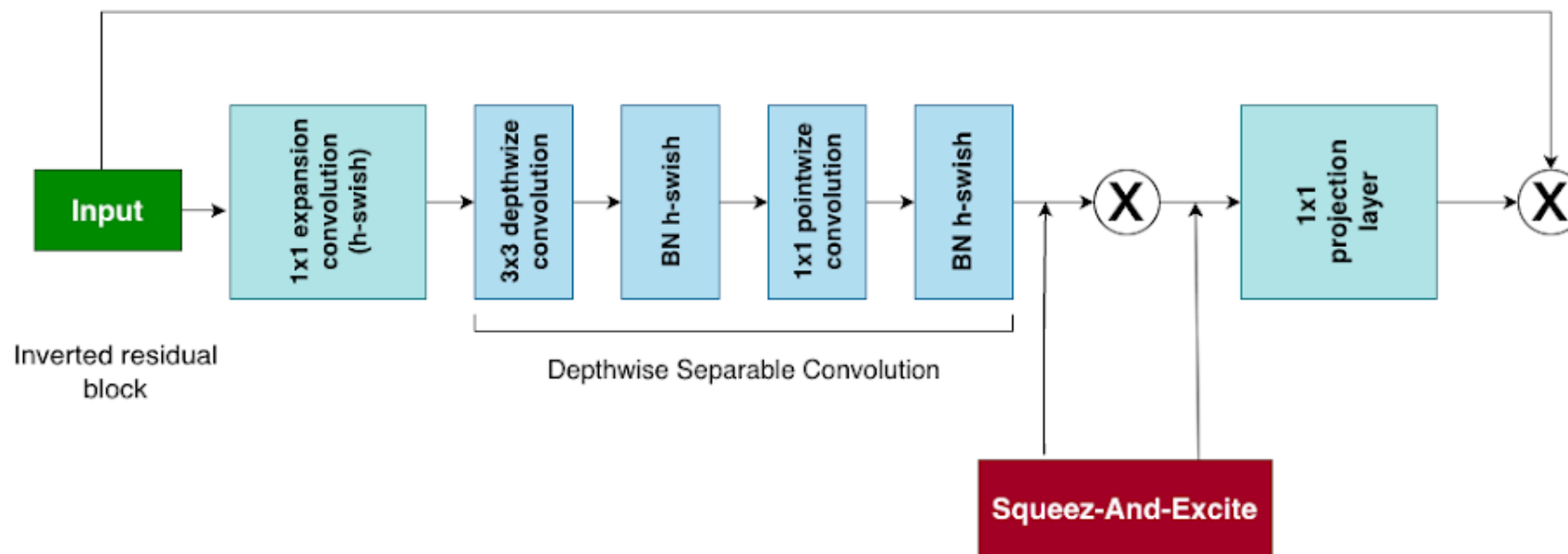


Figure 1.6: MobileNetV23 Architecture
Mifthauddin et al.2025[36]]

Methodology

Model Selection

II) Transfer Learning Models:

iii) DenseNet121:

DenseNet121 is a convolutional net architecture that combines dense skip connections between all the layers. This design allows the feature maps to travel smoothly across the network, which accelerates gradient propagation and reduces the probability of the disappearance of the gradient during training. DenseNet121 promotes feature reuse, whereby earlier learned representations can be utilized by later layers. It is smaller in size and has fewer parameters than other deep architectures. DenseNet121 particularly performs well in tasks such as locating coconut disease since it is able to retain small information in leaf patterns, textures, and laceration forms. DenseNet121 is an effective backbone model in a variety of agricultural image classification tasks since it is efficient and able to generalize. A DenseNet121 architecture is shown in Figure 3.6 [34].

Methodology

Model Visualization

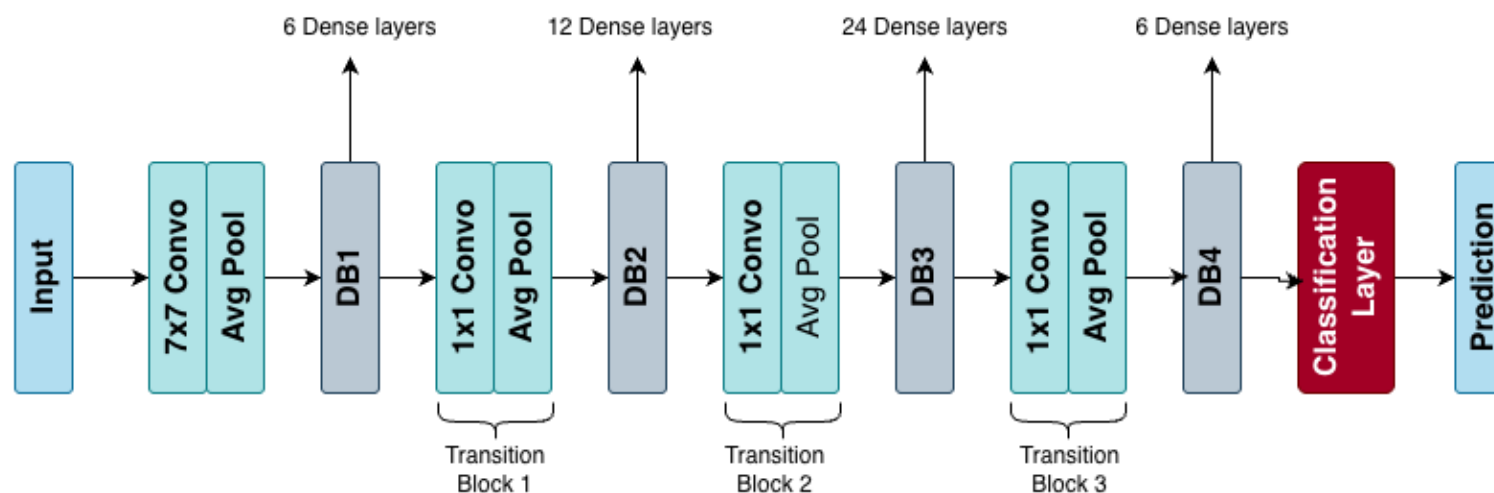


Figure 1.7: DenseNet121 Architecture. Diagram
Mifthauddin, et al. 2025[34]

Methodology

Model Selection

II) Transfer Learning Models:

iv) ResNet50

The ResNet50 training system uses residual learning, which allows deep networks to be trained without vanishing gradients. It has proven to be efficient in bio-image and natural image classification in various studies, such as the classification of plant and crop species [35]. This study uses ResNet50 to assess the effectiveness of the skip-connection network in detecting the specific structural clues evident among various species of flowers.

Methodology

Model Visualization

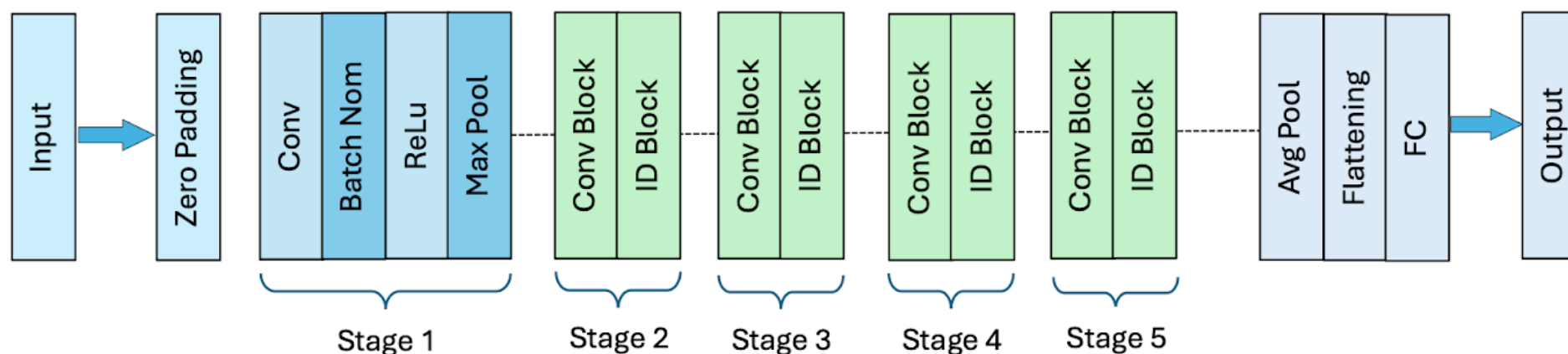


Figure 1.8: ResNet50 Architecture Diagram S.
Mukherjee et al.2022[35]

Methodology

Model Selection

II) Transfer Learning Models:

v) Xception

Xception is a neural network that applies depth-wise separable convolution to learn very thin discriminative visual patterns. Previous research indicates that Xception performs better than conventional CNNs in the intricate image classification problems, such as flower and botanical [10][15]. It is best suited to extracting fine-grained patterns of species with similar colour tones or petal shapes. Xception is used in this work as a high-performance baseline in the assessment of an advanced model's accuracy.

Methodology

Model Visualization

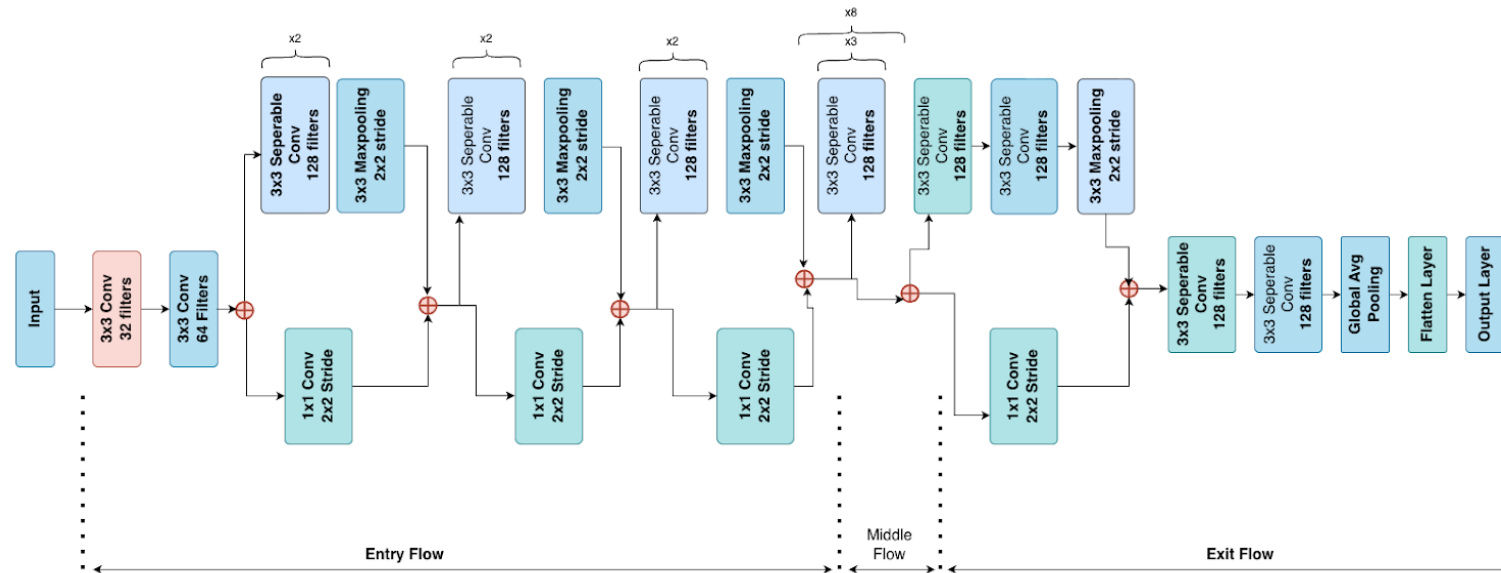


Figure 1.9: Xception Architecture Diagram K.
Srinivasan et al.2021[37]

Methodology

Performance Evaluation Metrics:

Accuracy, Precision, Recall, and F1-Score were computed to validate classification effectiveness. In this instance, we figure out the job's efficiency using accuracy, precision, recall, and the F1-Score. This four-factor paradigm is quite helpful for analyzing predictive data.

- ❑ **Accuracy:** Correct predictions over total predictions $\rightarrow \frac{(TP+TN)}{(TP+FP+FN+TN)}$
- ❑ **Precision:** Correctly predicted positives over total predicted positives $\rightarrow \frac{TP}{(TP+FP)}$
- ❑ **Recall:** Correctly predicted positives over all actual positives $\rightarrow \frac{TP}{(TP+FN)}$
- ❑ **F1-Score:** Harmonic mean of Precision and Recall $\rightarrow 2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$

Methodology

XAI Implementation (Explainable AI)

- Added XAI to interpret how the model makes decisions, reducing the “black-box” nature of deep learning.
- Helps users see which parts of the image influenced the prediction.

Methodology

XAI Techniques

❑ Grad-CAM

- Generates heatmaps highlighting important flower regions.
- Most accurate & meaningful visualization among all methods.

❑ Grad-CAM++

- Provides finer, more detailed highlights.
- Good localization but slightly less consistent than Grad-CAM.

❑ LIME

- Explains prediction by perturbing image segments.
- Useful but produces more scattered highlights.

Result and Analysis

Model	Precision	Recall	F1-Score	Accuracy
MobileNetV2	0.99	0.99	0.99	98.70%
MobileNetV3	0.98	0.98	0.98	97.60%
DenseNet121	0.99	0.99	0.99	99.99%
ResNet50	1.00	1.00	1.00	99.98%
Xception	0.99	0.99	0.99	98.69%

Table 1.2.: Result Table

Result and Analysis

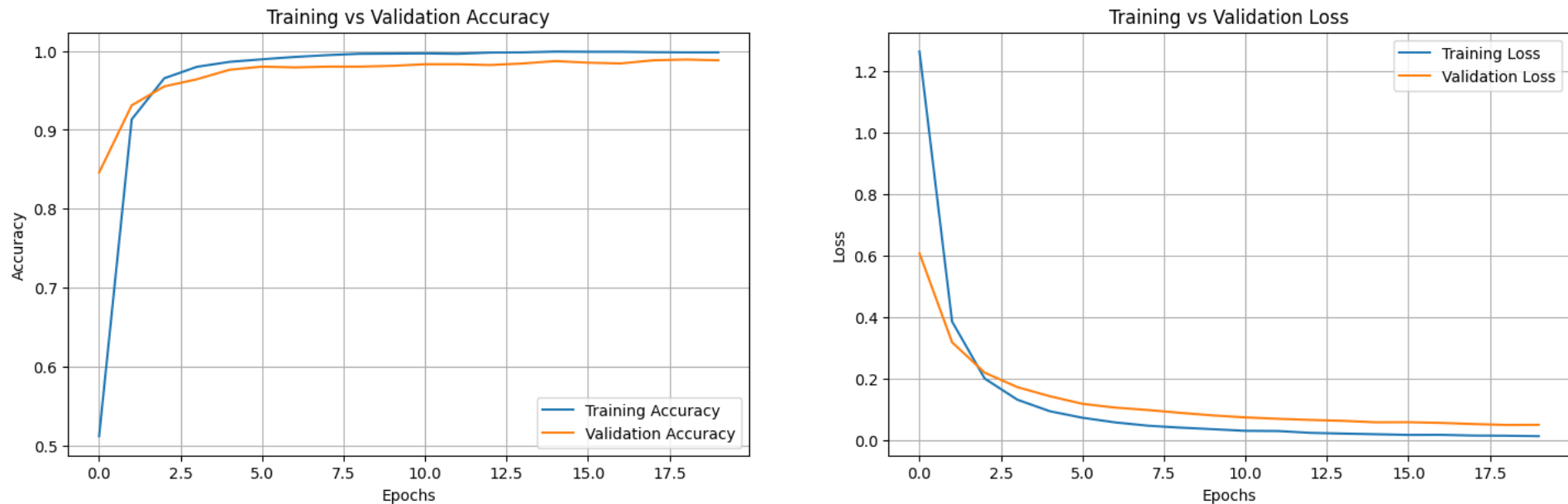


Figure 2.1(i): Accuracy & Loss Curve MobileNetV2

Result and Analysis

Class	Precision	Recall	F1-Score	Support
Bud_Rot	0.96	1.00	0.98	200
Coconut_Caterpillar	1.00	1.00	1.00	200
Leaf_Rot	1.00	0.94	0.97	200
Stem_Bleeding	1.00	1.00	1.00	200
Yellow_Leaf_Disease	0.99	1.00	0.99	200
Accuracy			0.99	1000
Macro avg	0.99	0.99	0.99	1000
Weighted avg	0.99	0.99	0.99	1000

Table 1.3: Classification report of MobileNetV2

Result and Analysis

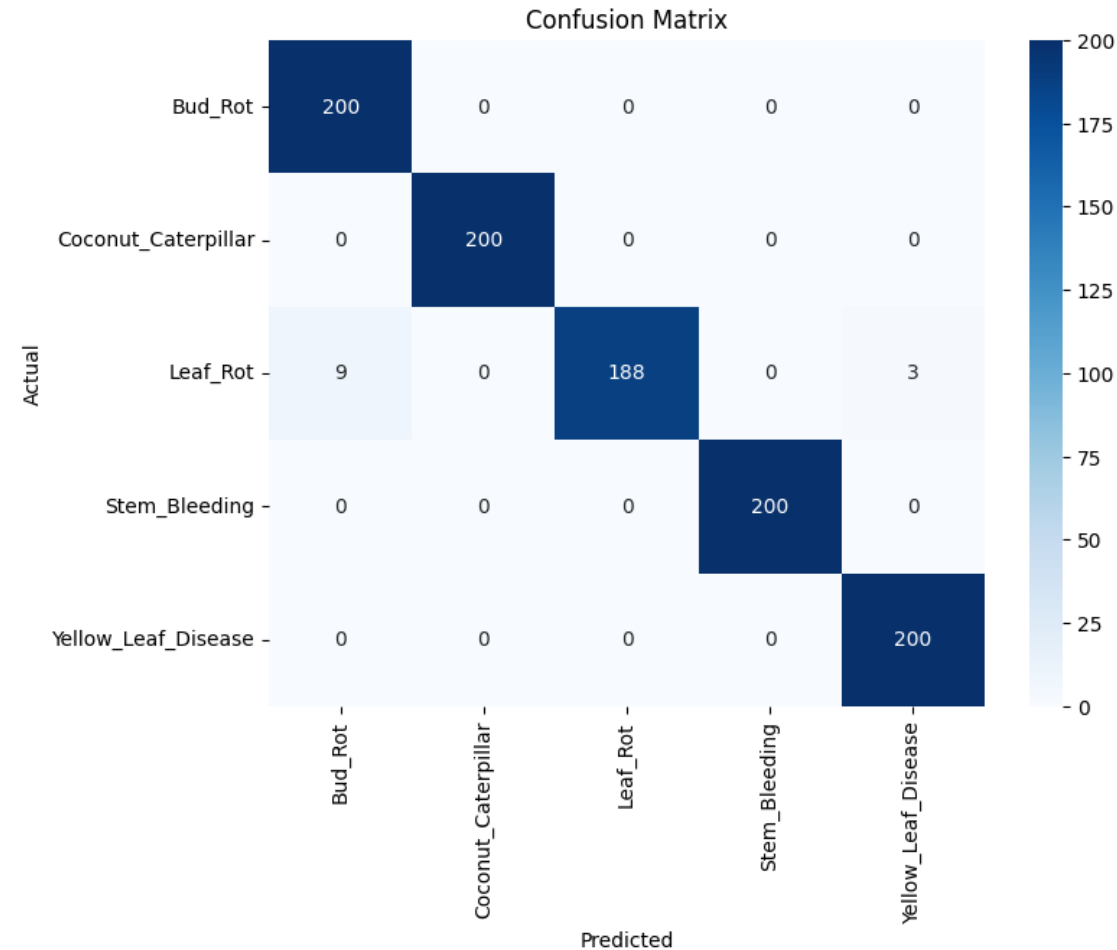


Figure 2.2(ii): Confusion Matrix MobileNetV2

Result and Analysis

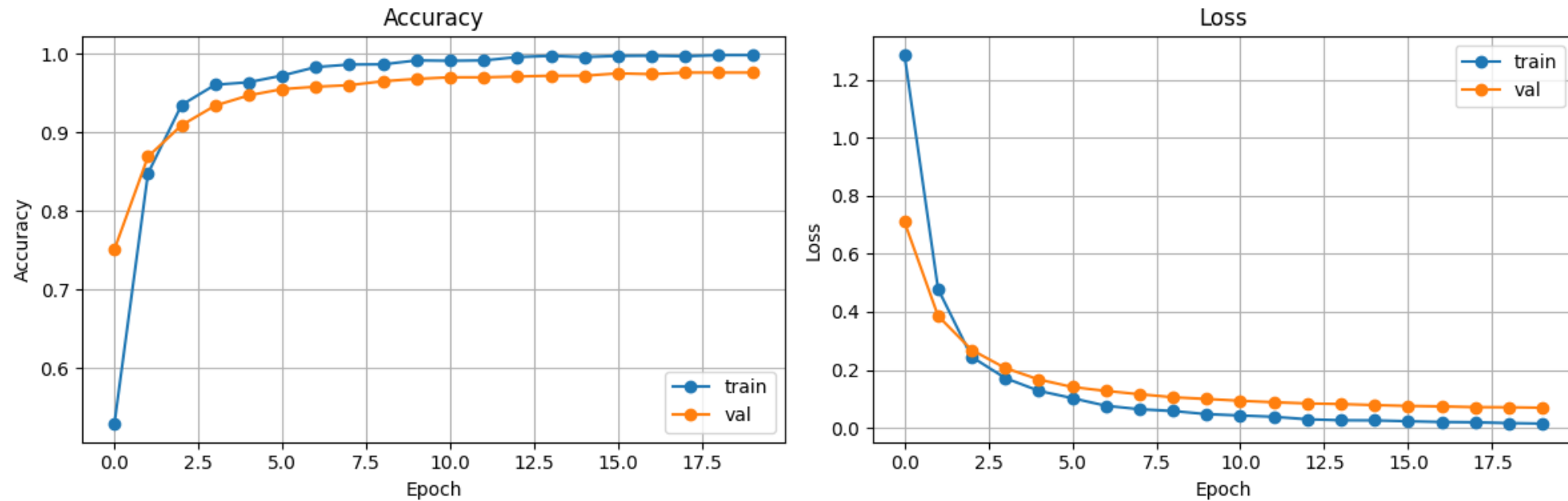


Figure 2.3(i): Accuracy & Loss Curve of MobileNetV3

Result and Analysis

Class	Precision	Recall	F1-Score	Support
Bud_Rot	1.00	1.00	1.00	200
Coconut_Caterpillar	0.96	1.00	0.98	200
Leaf_Rot	0.99	0.93	0.96	200
Stem_Bleeding	1.00	0.95	0.97	200
Yellow_Leaf_Disease	0.93	1.00	0.97	200
Accuracy			0.98	1000
Macro avg	0.98	0.98	0.98	1000
Weighted avg	0.98	0.98	0.98	1000

Table 1.4: Classification report of MobileNetV3

Result and Analysis

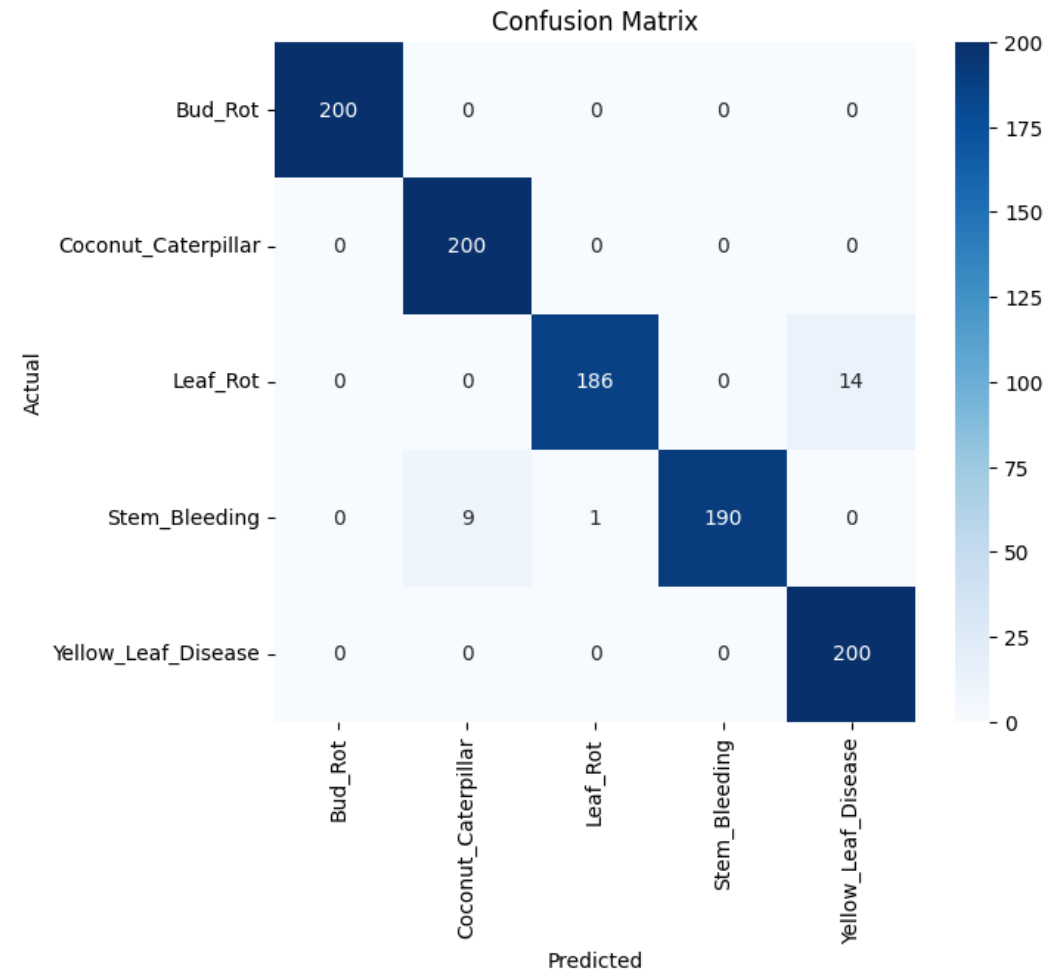


Figure 2.4(ii): Confusion Matrix (MobileNetV3)

Result and Analysis

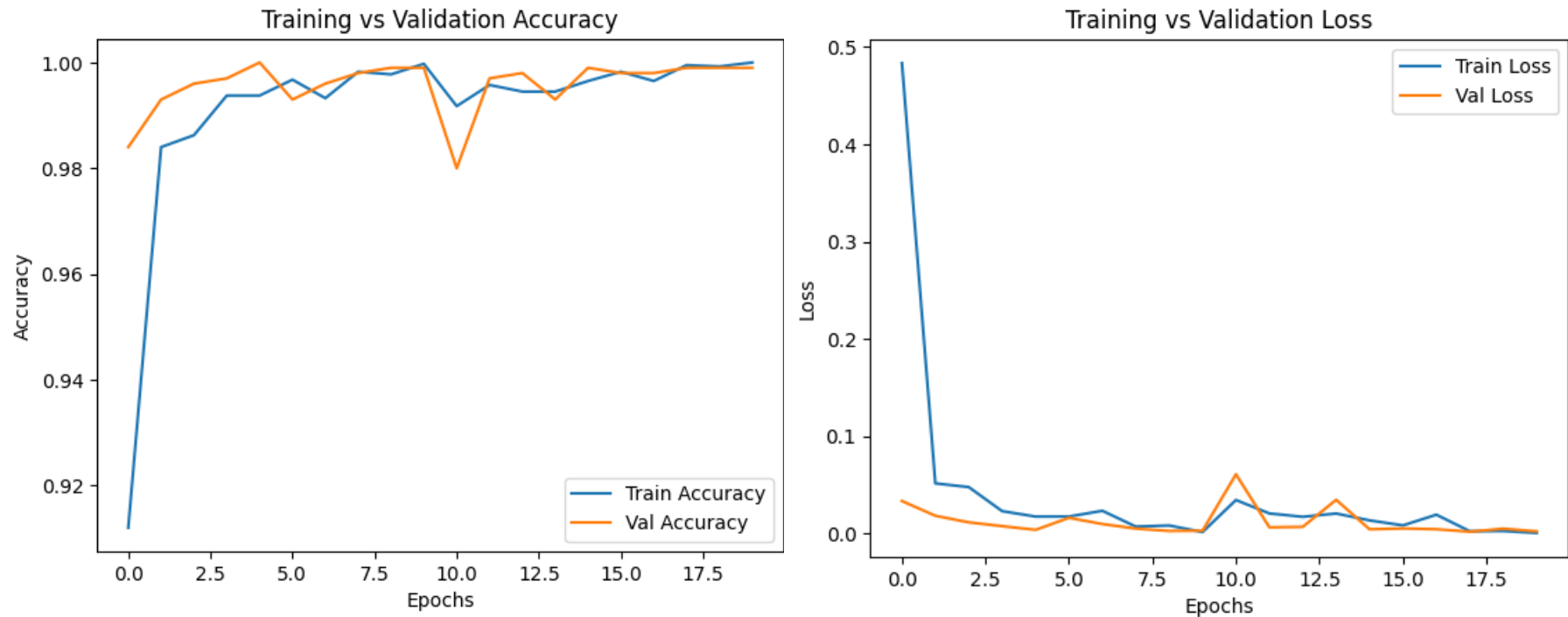


Figure 2.5(i): Accuracy & Loss Curve (DenseNet121)

Result and Analysis

Class	Precision	Recall	F1-Score	Support
Bud_Rot	0.99	1.00	1.00	189
Coconut_Caterpillar	1.00	1.00	1.00	191
Leaf_Rot	1.00	1.00	1.00	213
Stem_Bleeding	1.00	1.00	1.00	204
Yellow_Leaf_Disease	1.00	1.00	1.00	203
Accuracy	1.00	1.00	1.00	1000
Macro avg	1.00	1.00	1.00	1000
Weighted avg	1.00	1.00	1.00	1000

Table 1.5: Classification report of DenseNet121

Result and Analysis

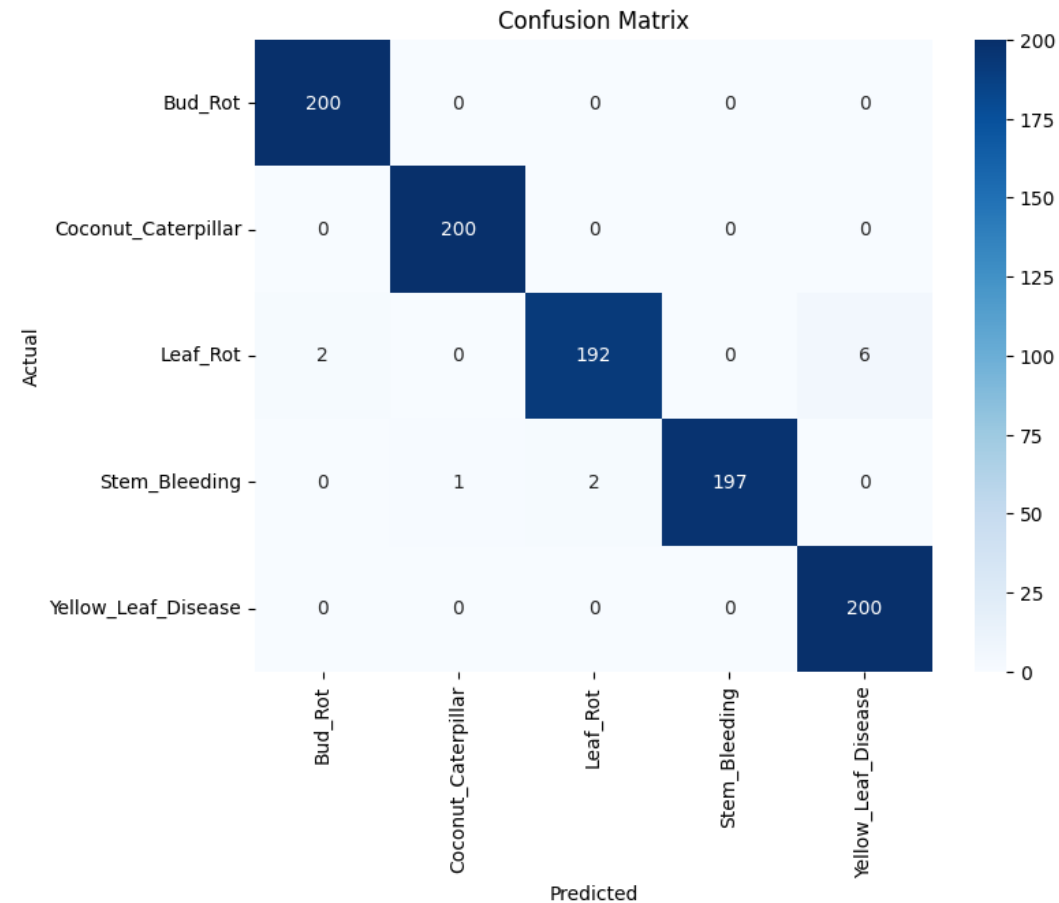


Figure 2.6(ii): Confusion Matrix (DenseNet121)

Result and Analysis

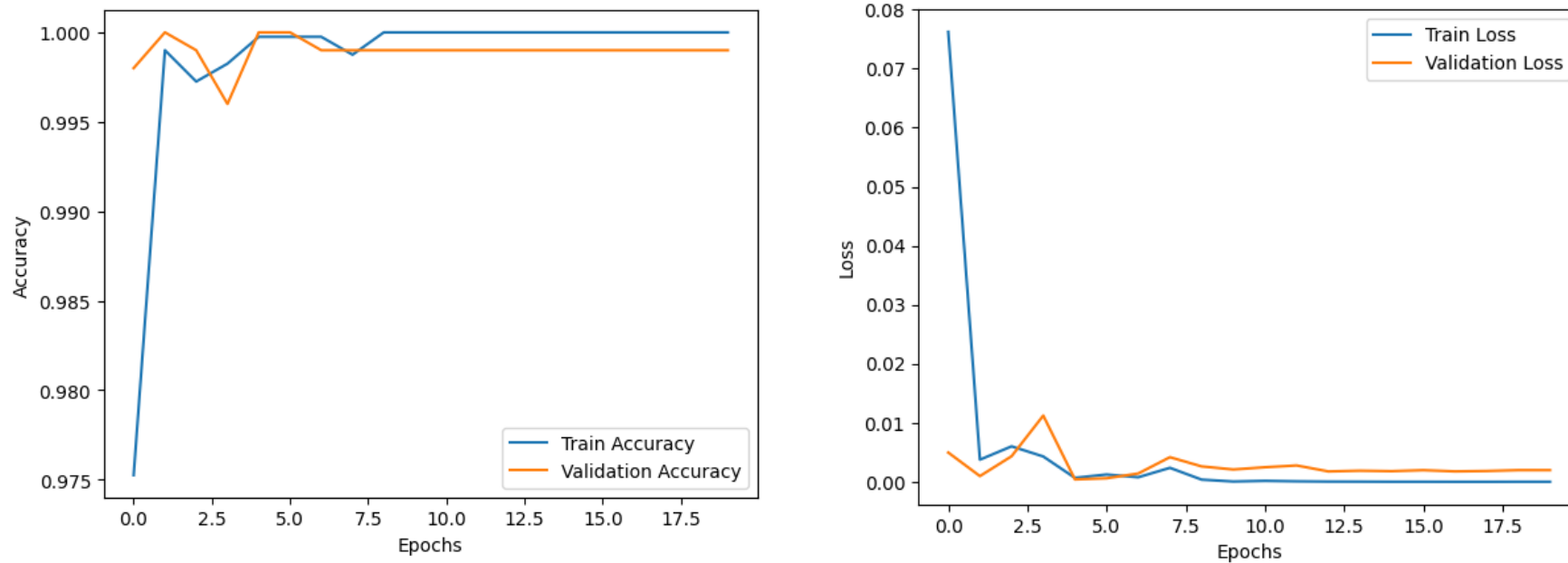


Figure 2.7(i): Accuracy & Loss Curve (ResNet50)

Result and Analysis

Class	Precision	Recall	F1-Score	Support
Bud_Rot	1.00	1.00	1.00	189
Coconut_Caterpillar	1.00	0.99	1.00	191
Leaf_Rot	1.00	1.00	1.00	213
Stem_Bleeding	1.00	1.00	1.00	204
Yellow_Leaf_Disease	1.00	1.00	1.00	203
Accuracy	1.00	1.00	1.00	1000
Macro avg	1.00	1.00	1.00	1000
Weighted avg	1.00	1.00	1.00	1000

Table 1.6: Classification report of (ResNet50)

Result and Analysis

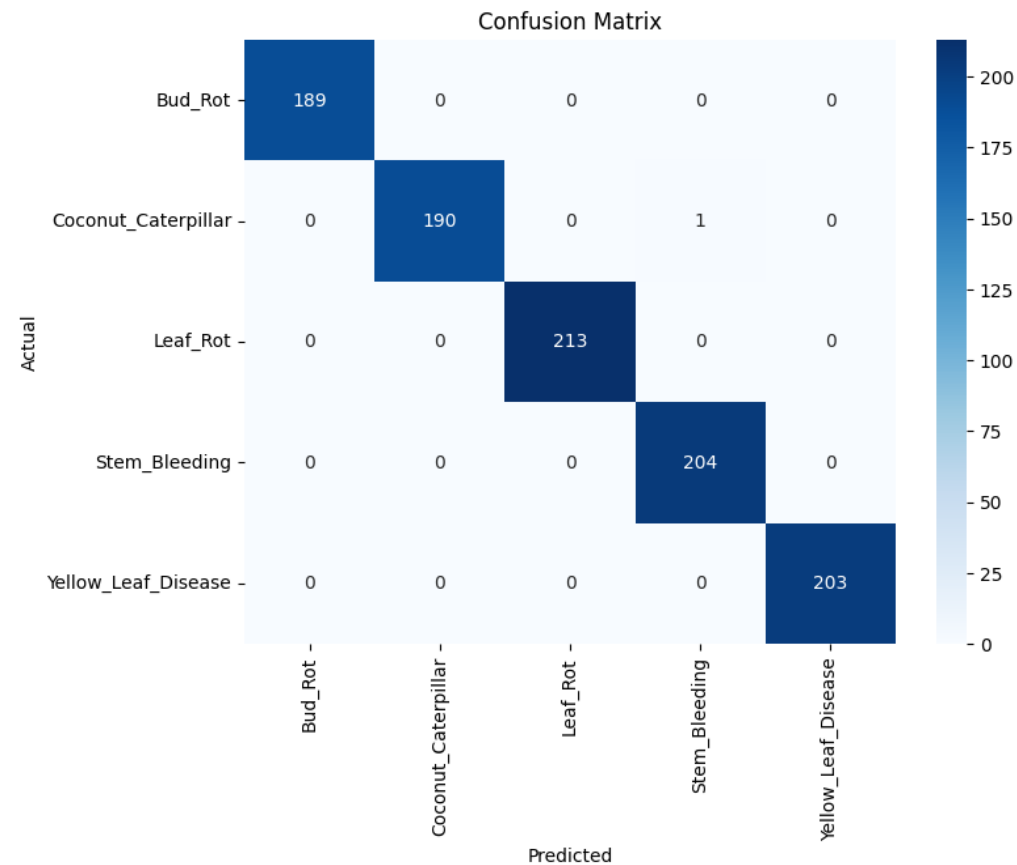


Figure 2.8(ii): Confusion Matrix of (ResNet50))

Result and Analysis

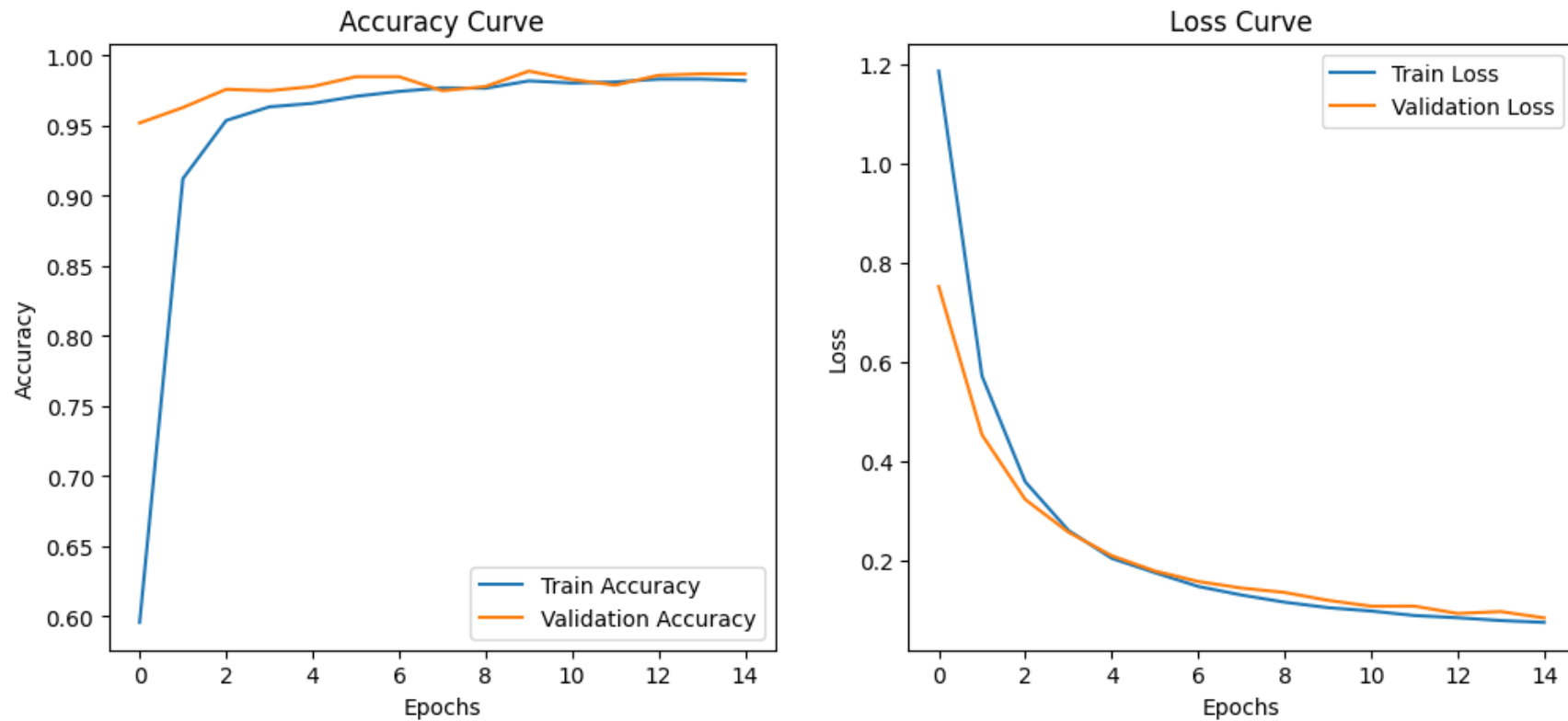


Figure 2.9(i): Accuracy & Loss Curve (Xception)

Result and Analysis

Class	Precision	Recall	F1-Score	Support
Bud_Rot	0.99	0.99	0.99	200
Coconut_Caterpillar	0.99	0.99	1.00	200
Leaf_Rot	0.98	0.99	0.99	200
Stem_Bleeding	1.00	0.97	0.99	200
Yellow_Leaf_Disease	1.00	1.00	1.00	200
Accuracy			0.99	1000
Macro avg	0.99	0.99	0.99	1000
Weighted avg	0.99	0.99	0.99	1000

Table 1.7: Classification report of (Xception)

Result and Analysis

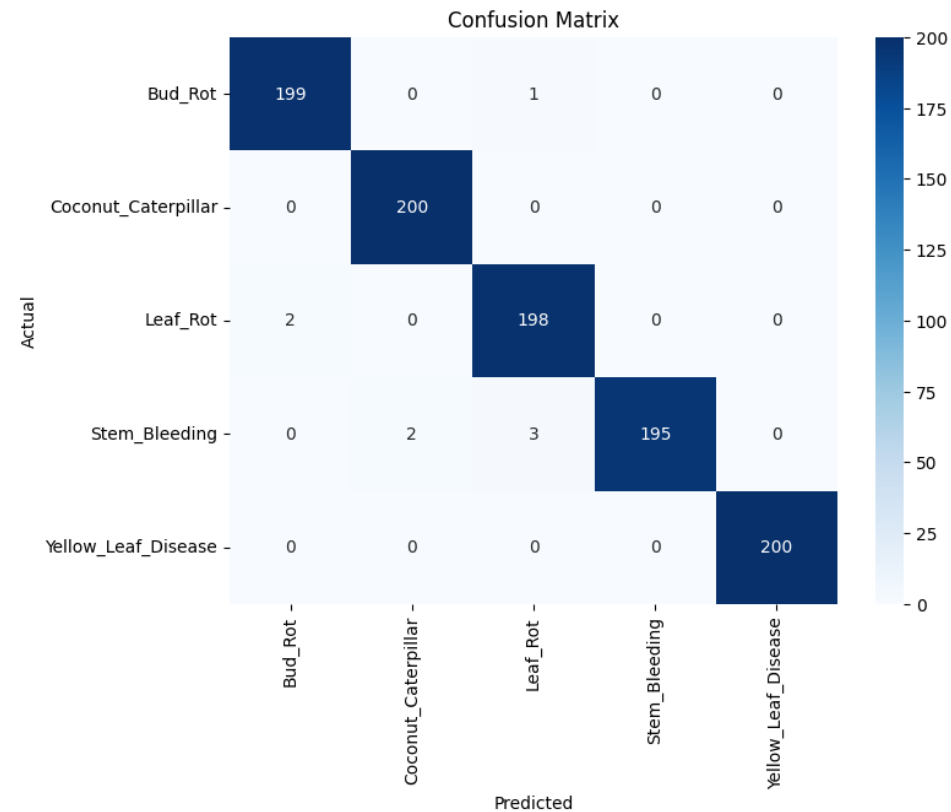


Figure 2.10(ii): Confusion Matrix (Xception)

Result and Analysis

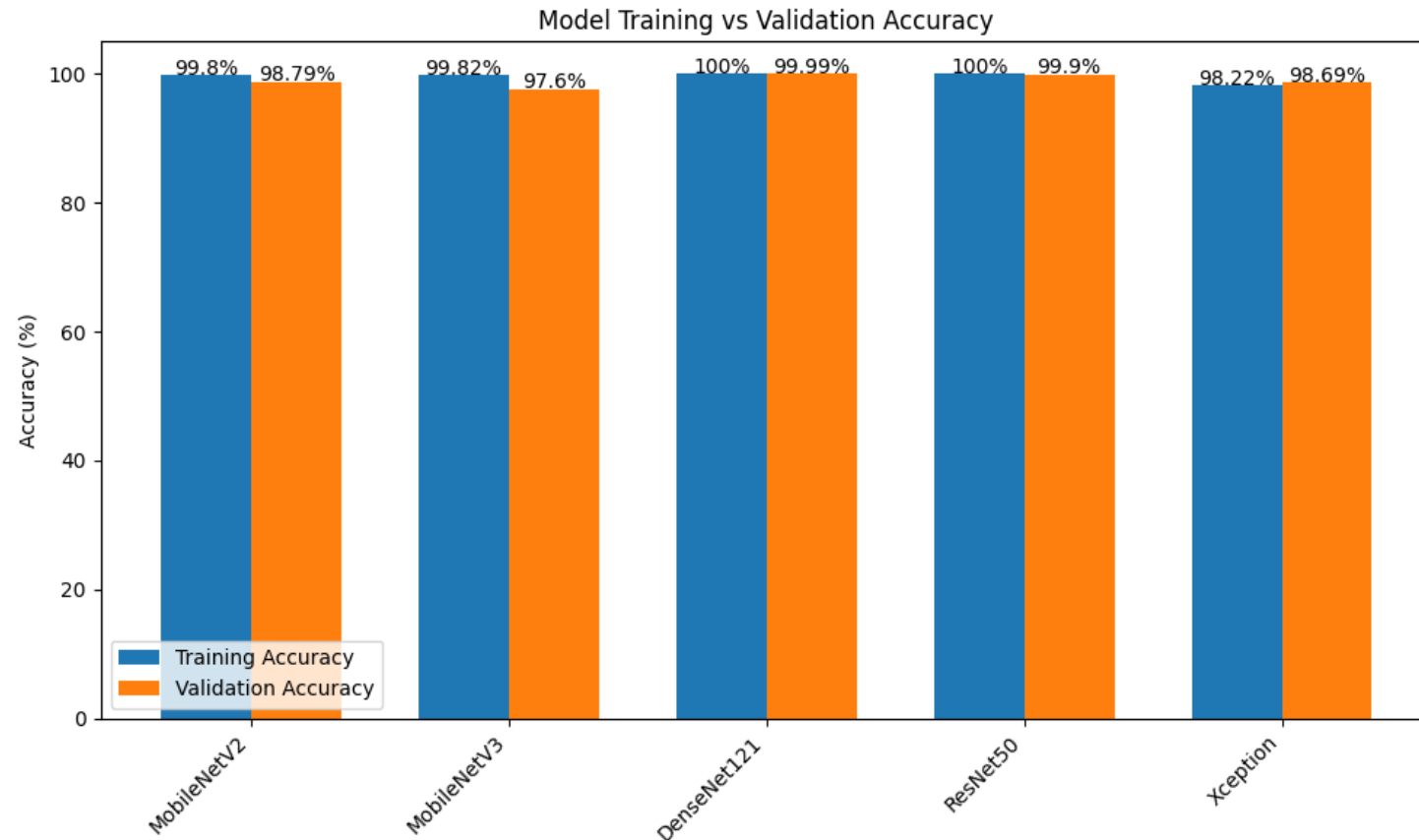


Figure 3.1: Comparison of accuracy between all models.

Result and Analysis

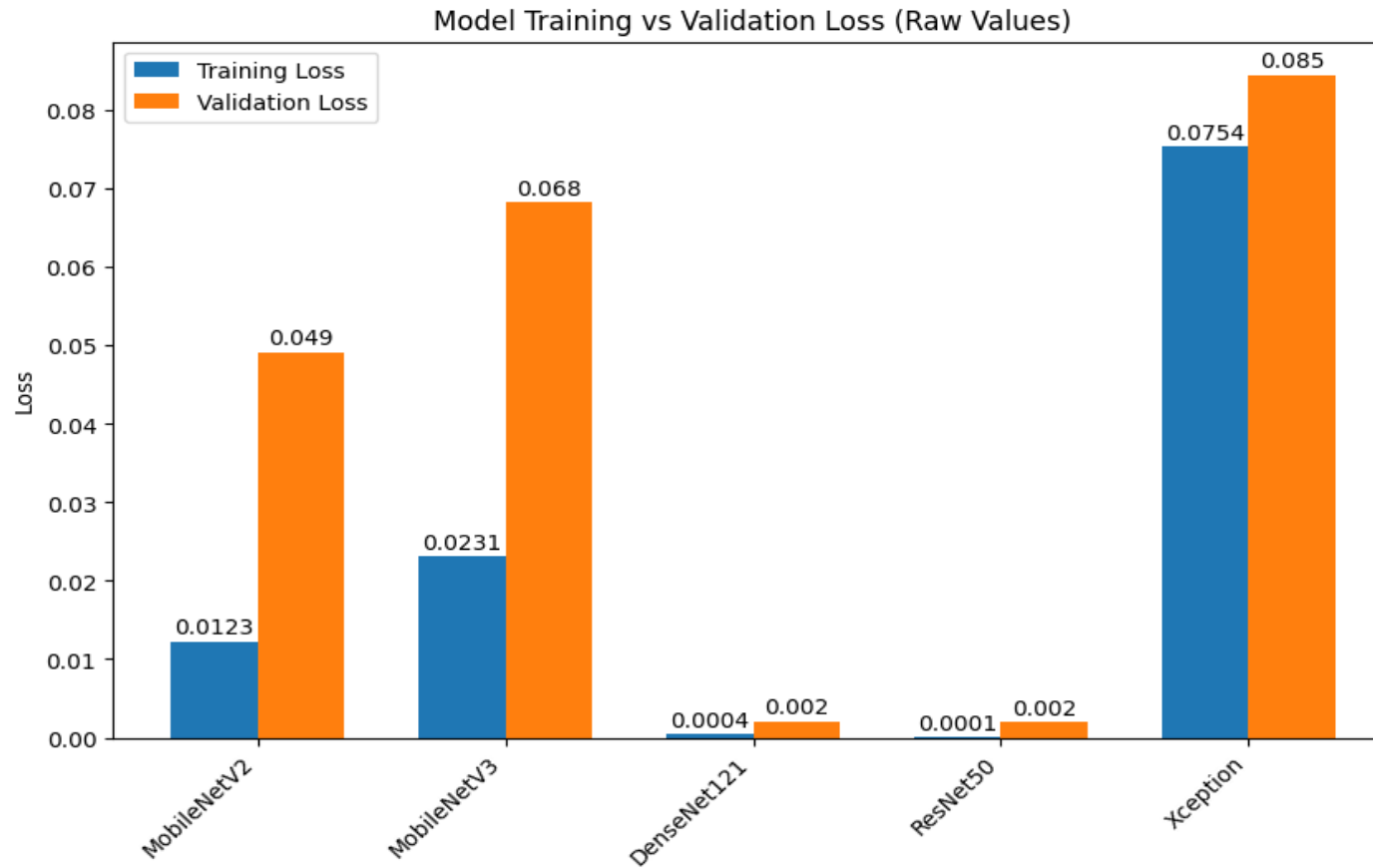


Figure 3.2: Comparison of loss between all models.

Result and Analysis

Explainable AI (XAI) Interpretation on MobileNetV2

❑ Grad-CAM

- Most accurate and stable visual explanations.
- Heatmaps focused on true disease features (patterns, structure, colour regions).
- Clearly distinguishes visually similar diseases (e.g., Leaf Rot vs Yellow Leaf).
- Shows MobileNetV2 learns real biological cues, not background noise.
- Reliable XAI method for this project.

Result and Analysis

Explainable AI (XAI) Interpretation on MobileNetV2

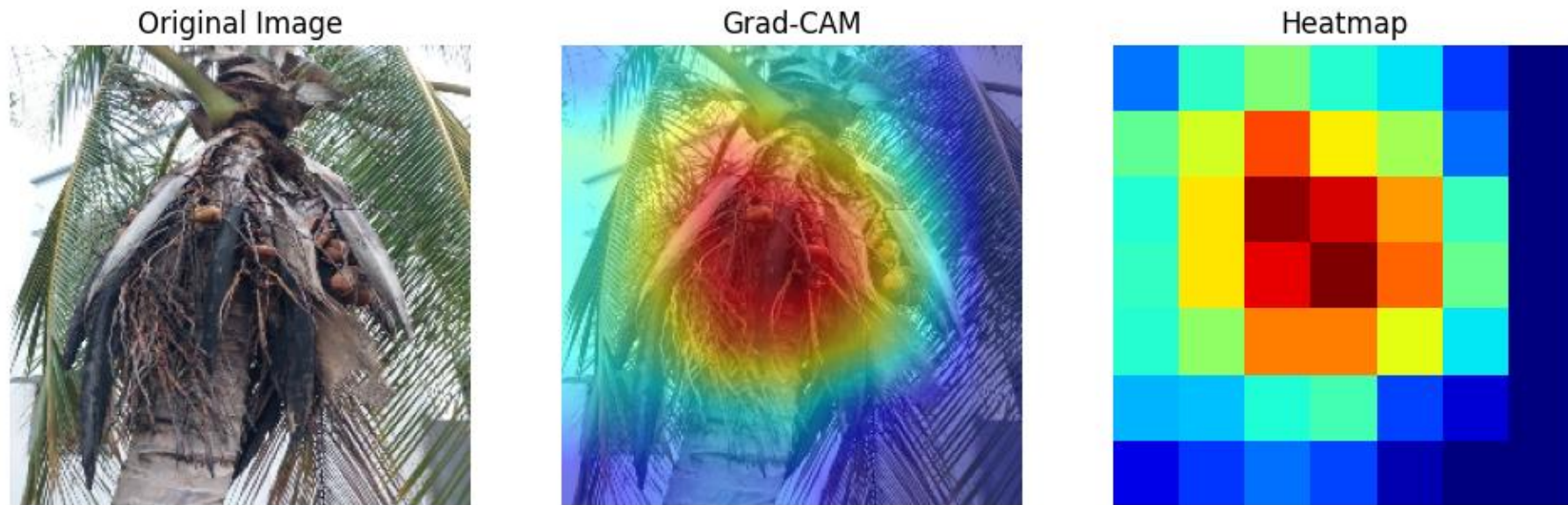


Figure 3.3: Grad-CAM Interpretation (On Best Performer)

Web App Deployment

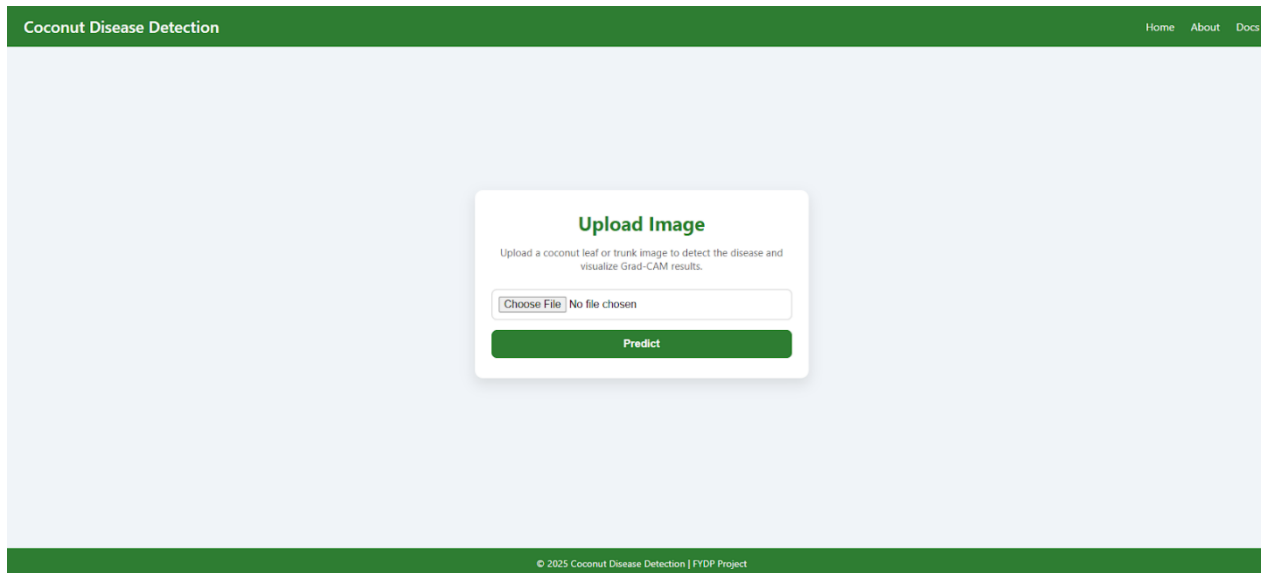


Figure 3.4: Web application home page.

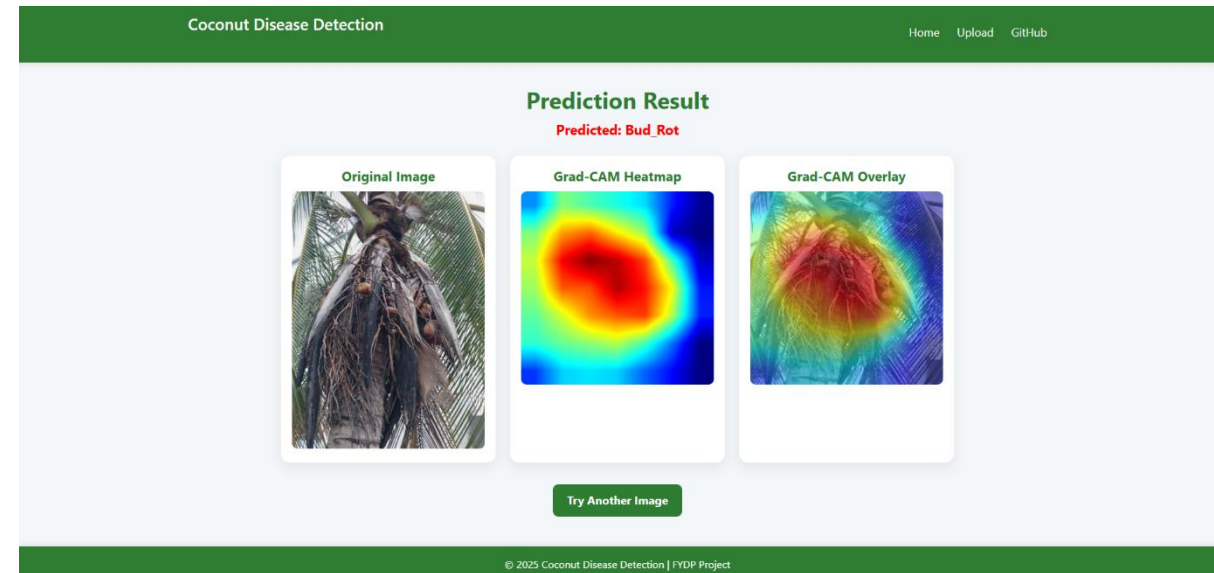


Figure 3.5: Web application result page.

Novelty of the Work

- ❑ Comparative evaluation of **5** deep learning models for coconut disease classification.
- ❑ Integration of XAI (Grad-CAM) for transparent and interpretable predictions.
- ❑ Identification of lightweight, high-accuracy models suitable for real-time use.

Conclusion

This research successfully developed an interpretable coconut disease detection system with a lightweight Transfer Learning model trained on a diverse dataset of 5000 images. MobileNetV2 emerged as the best-performing architecture, achieving 98.79% accuracy while remaining lightweight and deployment-friendly. The integration of the XAI technique Grad-CAM enhanced model transparency by revealing the visual cues used during prediction. The system was further implemented through a web application, demonstrating practical real-time usability. Overall, the project delivers an accurate, reliable, and explainable solution for automated coconut disease identification.

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A Deep Learning Approach for the Coconut Tree Disease Detection

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